

- 3 Therapeutic proteins are used to treat disease. One example of a therapeutic protein is human growth hormone (hGH).

hGH has important roles in growth during childhood and in regulation of metabolism in adulthood.

Children described as hGH-deficient do not produce enough hGH and grow more slowly than other children. People who were hGH-deficient when they were children have a mean adult height that is 32cm shorter than the population mean.

Daily injections of hGH are a treatment for hGH-deficient children that can increase growth rate, resulting in an increased adult height.

(a) The growth rate of three children was measured.

- The growth rate of the hGH-deficient child who did **not** receive daily hGH injections was 2.5 cm year^{-1} .
- The growth rate of the hGH-deficient child who received daily hGH injections was $10.0 \text{ cm year}^{-1}$ in the first year after starting treatment.
- The growth rate of the child who is **not** hGH-deficient was 5.0 cm year^{-1} .

Calculate the percentage increase in growth rate of the hGH-deficient child treated with hGH injections compared to the child who is **not** hGH-deficient.

Show your working.

..... % [2]

When this form of treatment started in 1958, hGH could only be obtained from the pituitary glands of people who had died.

In 1981, a plasmid containing hGH cDNA was constructed and inserted into *Escherichia coli* bacteria. This allowed recombinant hGH protein to be produced by the bacteria.

(b) Name the type of enzyme that:

- cuts plasmid DNA

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- makes cDNA from hGH mRNA.

..... [2]

- (c) Identify **and** explain **two** properties of plasmids that allow them to be used as vectors of hGH cDNA into cells of *Escherichia coli*.

1.....
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2.....
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..... [2]

- (d) In 1985, several cases of a rare brain disease were discovered in people who had been treated many years previously with hGH obtained from pituitary glands. It was decided, from 1985 onwards, that only recombinant hGH should be used to treat patients.

Explain the advantages of producing human therapeutic proteins, such as hGH, by recombinant DNA technology.

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- (e) hGH is one of many proteins in the body whose secretion or production is controlled by a person's sleep-wake pattern.

The sleep-wake pattern describes when, during a 24 hour day, a person is asleep and when they are awake. For example:

- pattern 1 – asleep during the night and awake during the day (normal)
- pattern 2 – asleep during the day and awake during the night.

Researchers used microarray analysis to identify which genes have their expression changed by a person's sleep-wake pattern. They collected mRNA from:

- a group of people with sleep-wake pattern 1
- the same group of people whose sleep-wake pattern was changed to pattern 2.

A summary of the results is shown in Table 3.1.

Table 3.1

sleep-wake pattern	number of genes with increased expression		
	during the day	during the night	all the time
pattern 1	661	733	108
pattern 2	134	95	8

- (i) Describe how changing the sleep-wake pattern from pattern 1 to pattern 2 affects the number of genes expressed.

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 [1]

- (ii) Explain how eukaryotic genes can be switched on and off, for example, at certain times of day.

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 [2]

- (iii) Explain how bioinformatics can help to identify whether the genes whose expression is changed by moving from pattern 1 to pattern 2 are important to health.

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 [2]

[Total: 15]